

(a) Resolution allocation

Denoising-evaluation budget

Native-HR

Target resolution $\times T$

Post-generation cascade

LR generation $\times T$

extra
SR / refine

$T + K$

InTraScale

LR prefix $\times s$

HR continuation
 $\times (T - s)$

T evaluations

■ low resolution ■ target resolution ■ extra process

Fixed total budget; no separate SR diffusion process

(b) InTraScale inference

Two learned roles connect a frozen LR prefix to a frozen HR continuation

